Case Study 1 - Atari

Student submission

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Worksheet link

https://courses.ethereallab.app/assignment/v4/IT265-001-F2026/case-study-1-atari/aac97

Grading link

https://courses.ethereallab.app/assignment/v4/IT265-001-F2026/case-study-1-atari/grading/aac97

AI disclosure: Completed; 1 entry; 1 declared AI use; 0 declared no AI use; policy v1

Game Overview

Game details

Response prompt

Identify the game: title, developer, publisher, original platform, genre, and release date. Note the version you played if different.

Submitted response

Title:Pac-Man (Atari 2600)
Developer: Atari, Tod Frye
Publisher: Atari, Sears
Original platform: Atari 2600
Genre: Action
Release: March, 1982
Citiation: https://atarionline.org/atari-2600/pac-man

History and industry

Response prompt

Explain the historical context of the release and where the game fit in the industry at that time.

Submitted response

Pac-Man launched on the Atari 2600 in March 1982, the home version of the large commercial boom of the arcade port. Atari secured the exclusive home console rights so they could drive the sales of the hardware. Atari produced roughly 12 million cartridges with the expectation of millions of customers buying the Atari console just to play the game. Pac-Man was a commercial success, selling over 7 million units and becoming Atari's best-selling game in history. This came at a cost because the massive change in quality from the arcade to the home console led to wide spread consumers dissapointment led to millions of unsold copies and returned copies. This damaged consumers' trust alongside the E.T. game the same year, which led to the North American video game crash of 1983.

https://en.wikipedia.org/wiki/Pac-Man_(Atari_2600_video_game)

Contemporary comparison

Response prompt

Compare the game with a similar game from the same period, explaining a meaningful design similarity or difference.

Submitted response

I'm gonna compare the original arcade to the Atari version based on everything. The original arcade had more life in it, with enemy ghosts being bright colors compared to the Atari version, with very grey colors and also blinking around the map, making it hard to see when eating a pellet. The Pac-Man in ht e ogrinal game is a yellow circle eating, but in the Atari version he has eyes showing a little more expression when eating and dying to a ghost. The movement on the arcade is better and more rewarding, while in the atari the pac man is hard to move up and the pellets are just like lines.

The noise it makes is also a sharp, horrible noise compared to the arcade machine, which is alot nice and more pleasing to the ears.

Gameplay Mechanics And Level Design

Core gameplay mechanics

Response prompt

Explain how the core actions and rules shape a recurring player decision. Support the explanation with a gameplay example.

Submitted response

The core loop of Pac-Man on the Atari is about collecting pellets to get points while avoidoing ghost the move around the map. The player must collect all the pellets scattered across the maze to clear the level and advance. There are four ghosts that roam around the maze. There are also four pellets that are placed at the corners of the amze the tempooraily make the ghost cosumable and move slower for the player to make contact with them and get more points to the board. The player is restricted to four movements up, down, left, and right, and they use a joystick to control it. This gives the players options when playing the game, whether they should trigger a power pellet early or later in the game while getting max points, making it high risk when it comes to point maximization. A game play example would be players move to the corner of the map to get the power pellet. Once comsumed the ghost on the map are start to flicker and change color and one ghost that is close gets eaten by the player and they turn into two eye balls and float back to the middle of the map to come back to there ghost form.

Level design

Response prompt

Explain how the layout, obstacles, or progression affect a player decision or challenge. Use a specific example.

Submitted response

The layout differs from the arcade version a lot, changing the way the player plays and moves around the map. Instead of the horizontal warp tunnels, they become vertical tunnels located at the top and bottom of the center of the screen. The layout of the map is the same on both sides, but they are hallways the conected to other hallways, and there built in a way that makes the player have to choose if they wanna commit to that hallway or move into another one. The hallways are cut into different parts, making it different throughout the gameplay. The pellets also help in the player's decision-making by making them choose if they should start moving to another corner or try to finish collecting one side completely before moving on. Progression across the rounds increases the ghost speed in moving around the map and cuts down the time duration of the power pellets. An example would be the player moving to the right, collecting wafers but realizing there's a ghost coming from the top of him, forcing him to either go back or move to a power pellet because of the layout of the map. The player has to decide where to go next while avoiding the ghost and trying to collect all the points to move on.

Graphic And Audio Design

Visual design

Response prompt

Explain how a visual design choice affects the experience or communicates information during play. Use a specific example.

Submitted response

Due to the hardware limitations of the Atari's Television interface adapter, which could natively display only two hardware player sprites per scanline, programmer Tod Frye cycled the rendering of the four ghosts across consecutive video frames. Rather than solid entities, the ghost appears on the screen only every fourth frame, making it look like a flicker effect. This impacted the gameplay because the ghost would constantly phase in and out of visibility, and it would be hard to tell which way they would go. This would also have an effect on the power pellets because sometimes you couldn't even tell if there able to be eaten with how they constantly flicker. This is from my own experience playing the

game and trying to get high points. An example is when I picked up the power and went to eat them without realizing it was over. Another clue is the audio, but I had it muted because the audio is harsh to the ears.

https://en.wikipedia.org/wiki/Pac-Man_(Atari_2600_video_game)
https://pacman.fandom.com/wiki/Pac-Man_(Atari_2600)

Sound design

Response prompt

Explain how music, sound effects, or their absence affect the experience or communicate information during play. Use a specific example.

Submitted response

There is no music in this game, there are only sound effects throughout the game. The absence of music can create a more focused environment on trying to beat the level and only hearing the sound effects of the game. The game has sound effects for picking up the pellets and the power, as well as eating the ghosts. The sound effects are very rough, with the eating of pellets sounds grating and not appealing to hear. The sound effect of eating the ghost has a nice jingle to it, making it rewarding to eat the ghost to hear that rather than the grain pellets. That probably wasn't the intent of the developer, but it gives more incentive to eat the ghost rather than just points. The boot-up sound of the game is loud and harsh to the ears. This is because of the game console hardware limitations to make clear sounds.An example would be me playing the game and turning off the sound because it's harsh and hard to focus when you have the grainy noise in your ears.

Narrative And Storytelling

Narrative and setting

Response prompt

Explain how the setting and story relate to play. Describe key events if present, or explain the effect of a limited or absent story.

Submitted response

There is no story ot and thing here that relates to the play. This is just bare bones gameplay of a yellow ball eating a ghost and pellets. There are no key events other than the power-ups letting you eat the ghost. This doesn't affect the gameplay whatsoever because this isn't a game to think about the story, but rather a game to just get high points.

Character development

Response prompt

Explain how character development relates to gameplay, or how limited or absent development affects the experience.

Submitted response

There isn't character development in the game at all, it's just a yellow circle eating ghosts and pellets. The absence of development doesn't affect the gameplay or experience at all. The ghosts do have names and are based on their colors, but in the Atari, you can't tell anybody apart. Pac-Man himself is all we get and doesn't change the experience at all.

Impact And Reception

Reception

Response prompt

Summarize the critical and player reception using identifiable reviews or feedback. Distinguish release-era reception from later reactions.

Submitted response

To summarize, the reception it got was mixed, with more of a critical review that came out for the version of the game. A review in Electronic Game Magazine talks about how Atari translated the core idea of the arcade and rules to home hardware decently but criticizes the presentation of it, pointing out the flickering ghost, the different map, and the dots

becoming dashes. This soured the marketing for this game from multiple people disliking this version of Pac-Man because it wasn't faithful to the original one. It sold millions, but it was estimated to make more, so they overproduced on the units, leaving millions of units unsold and also getting a high return rate. This is what led to the downfall of the game industry in America. Later, they released a version of the Atari with the old game, and now in this current age it's looked back at fondly and has good reviews mainly because of the special edition console that came with the rerelease.

Citation:https://intotheverticalblank.com/2017/04/05/pac-man-electronic-games-magazine-and-the-exact-moment-atari-lost-the-video-game-war/

Industry and cultural impact

Response prompt

Explain the game's industry impact and cultural significance using research. Avoid claiming influence that the sources do not establish.

Submitted response

The game industry and cultural significance of this game are large. This game became the best-selling home cartridge in the console's lifespan, but it was also the catalyst for the massive market collapse known as the North American video game crash of 1983. The overproduction of the Pac-Man cartridges was one of the leading causes of its downfall and the market collapse. Over 5 million copies didn't sell even after it went on discount, and with the launch of ET, retailers noted that customers weren't buying the game and noted it was not a worthwhile product to have in stores. The business model of Pac-Man was to piggyback on the arcade success, but without it exceeding expectations, the arcade people didn't want it, and this can be traced back to Atari's financial downfall. Warner Communications reported massive revenue shortfalls by December 1982, which led to an industry-wide panic that forced smaller third-party developers into bankruptcy and opened the door for Nintendo to perform well in the American game industry. When it comes to the cultural impact of pac man atari there is no other article talking about the Atari version other than that it led to the downfall of the game industry at that time. Articles talking about the original Pac-Man arcade game and its impact. But one thing is certain the Pac-Man Atari version showed limitations of a home console compared to the arcade counterpart. This showed consumers the limitations while showing what it may be capable of, with Nintendo's home console showing how it can improve and become better.

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Reflection

Fun and experience

Response prompt

Explain what made the game enjoyable or frustrating for you, using a gameplay example.

Submitted response

It was a mix of both for me when it came to play. Its pac man the core idea and gameplay are there, so it's gonna be enjoyable to me no matter what. I've always enjoyed playing Pac-Man, so this just felt like another version of the game. That was my enjoyment of the game, but the frustration was the controls and the style of it. The flickering ghost is the most frustrating part of it, with it being hard to tell when they're able to be eaten and everything. That and the audio was not fun, so I had it on mute to play the game. It also felt unresponsive, and in moments, especially when I'm trying to move up, sometimes I would tend to go back and forth all the time because it wants you to be precise in that movement, so I get stuck leaving the corner of the map. Over all thought my enjoyment of the game was alright to me; I prefer the original or even the Google version.

Learning and research

Response prompt

Reflect on what this case study taught you and how easy or difficult it was to find dependable information.

Submitted response

This case study wasn't easy because it was hard to find information about this type of Pac-Man version. Pac-Man is so massive a video game icon that when it came to searching anything about Pac-Man, it always pointed me to the original arcade version of it. Even when I tried searching for the Atari version, it would be hard because Atari re-released the console with the game to make a special edition. Every time I searched that up, it would just point me to that console page to look at and buy. Wiki helped a lot with my research because it provided lots of information about that type of version and game I was looking for. This taught me about the video game industry crash in America, which I didn't know much about before researching this, so it taught me the troubles of the industry and how it could help even during this time of the game industry, and how it's struggling to find a footing as of late, with massive companies doing layoffs but also helping indie developers rise to the occasion. I liked learning about this, and it taught me a lot.